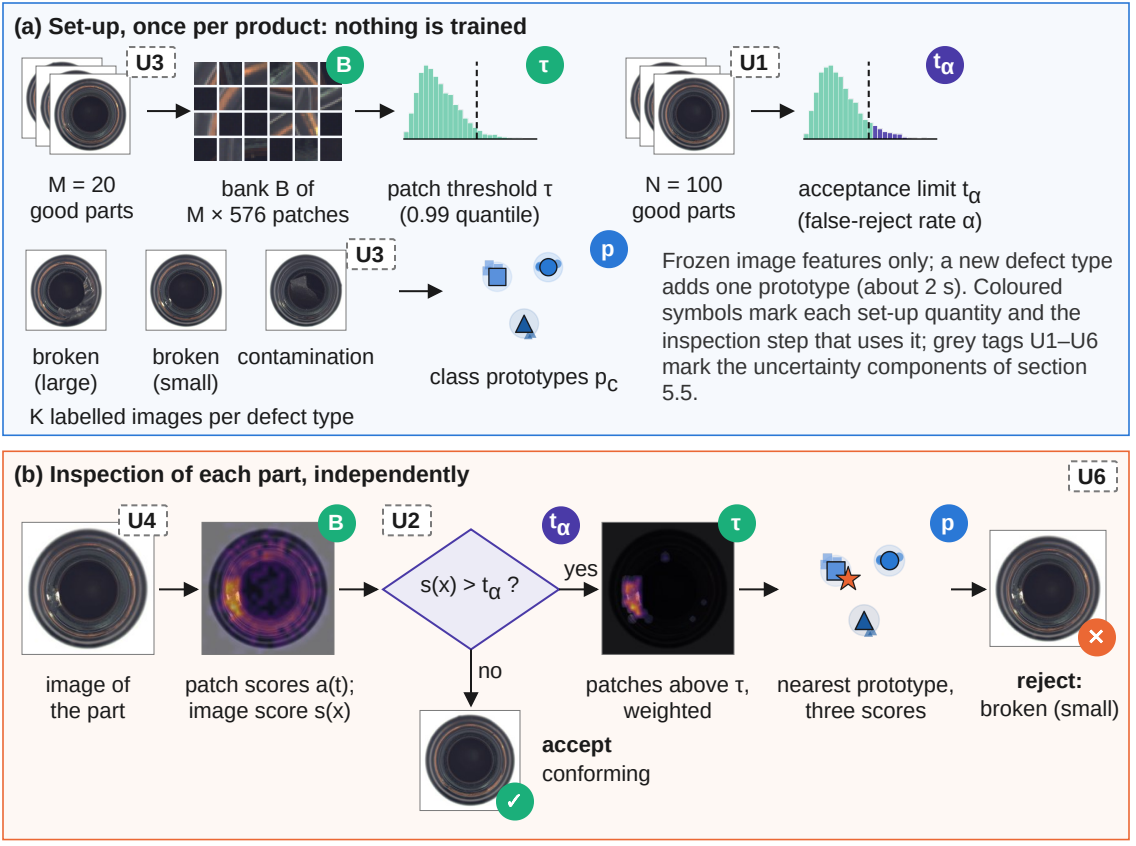

(a) Set-up, once per product: nothing is trained
U3
B
τ
U1
t_α
M = 20 good parts
bank B of M × 576 patches
patch threshold τ (0.99 quantile)
N = 100 good parts
acceptance limit t_α (false-reject rate α)
U3
p
broken (large)
broken (small)
contamination
K labelled images per defect type
class prototypes p_c
Frozen image features only; a new defect type adds one prototype (about 2 s). Coloured symbols mark each set-up quantity and the inspection step that uses it; grey tags U1–U6 mark the uncertainty components of section 5.5.
(b) Inspection of each part, independently
U6
U4
B
U2
t_α
τ
p
s(x) > t_α ?
yes
no
image of the part
patch scores a(t); image score s(x)
patches above τ, weighted
nearest prototype, three scores
reject: broken (small)
accept conforming